



# Explainability driven post-hoc correction enhancing deep neural network performance for job cycle time prediction

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## Abstract

Deep learning models promise a lot for predicting job cycle times in manufacturing but they often fall short when working with the limited data sets typical of real-world factories. To tackle this generalization problem, we have developed a post-hoc correction framework that uses SHAP explainability to make smarter adjustments. Instead of applying a blanket fix our method first pinpoints which new jobs are likely to have inaccurate predictions. Then for only those jobs it uses insights from SHAP to fine-tune the most critical input attributes before feeding them back into the original unchanged deep neural network (DNN). When we tested this on a real 300-job data set from a wafer fab, the results were striking the framework slashed the Root Mean Square Error (RMSE) by 22.5%. This correction brought the DNN's performance to a level that was better than or just as good as standard benchmarks like Random Forests showing it's a practical way to get more out of deep learning models even when data is scarce.

**Keywords** Post-hoc corrective framework · Model-agnostic · Explainable artificial intelligence · Deep neural network · Cycle time prediction

## 1 Introduction

In modern manufacturing the ability to accurately predict job cycle time is not merely beneficial—it is critical for operational excellence [1]. These predictions form the backbone of key managerial functions from production planning and due date assignment to dynamic scheduling and resource allocation [2]. However, achieving accurate predictions is notoriously difficult. The factory floor is a dynamic environment fraught with challenges such as irregular job releases scheduling deviations human variability and diverse product mixes that complicate forecasting efforts [3]. To navigate this complexity, researchers have increasingly turned to machine learning (ML) and deep learning (DL) methods [2–15].

To enhance the effectiveness of these methods for job cycle time prediction, several treatments have been taken in past research. The first is to adopt job cycle time prediction

models with deeper architectures such as feedforward neural networks (FNNs) with multiple hidden layers [9, 11, 15] two-dimensional long short-term memory models (BiLSTMs) [6] stacked sparse autoencoder-FNNs (SSA-FNNs) [8] convolutional neural networks (CNNs) [14] etc. The second is to add a preprocessing mechanism that selects converts or merges relevant attributes [5, 6, 9]. The third is to add a job classification mechanism or form an ensemble of prediction methods [2, 3, 7, 11, 13]. The fourth is to gather/consider more inputs using Industry 4.0 or big data analytics techniques such as cyber physical systems RFIDs and autoencoders [5, 7, 12, 16].

Another possible approach is to add an attention mechanism [17] to the prediction model, which is a hot topic in explainable artificial intelligence (XAI) [18–22] and has shown its effectiveness in various pattern recognition [23–26] and time series prediction problems [16, 27–34] but is rarely used in job cycle time prediction. Although job cycle time can be viewed as a time series it is best viewed as an input–output relationship for the following reasons [1, 35, 36]:

- A factory is not a closed system and may undergo rapid and unexpected changes [37].

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- Different types of products are manufactured in a factory at the same time [38, 39]. Therefore, the cycle times of jobs for the same product may not constitute a continuous time series.

Therefore, a cycle time prediction problem does not require deep neural networks for time series prediction. In addition, the Post-hoc Corrective Framework suitable for a job cycle time prediction problem is also different from the existing attention mechanisms for time series prediction problems.

To further enhance the effectiveness of job cycle time prediction using a deep neural network (DNN) this study establishes a SHAP-driven post-hoc correction framework. Our objective is to address the prevalent issues of poor generalization and overfitting that DNNs often encounter in complex data-limited environments like wafer fabrication. The novelties and main contributions of the proposed methodology are as follows:

- The correction framework is uniquely designed for input–output relationship prediction differentiating it from typical attention mechanisms used in time series analysis [27–34].
- Rather than applying a blanket correction to all new data the framework selectively and specifically targets individual job instances that are preliminarily identified as likely to have inaccurate predictions [27–34].
- Using a SHAP analysis our mechanism identifies the most critical attributes for a single prediction and performs a precise localized correction based on this insight. This concept is inspired by attention mechanisms used in neural machine translation [40].
- We provide rigorous benchmark comparisons and statistical validation to demonstrate that our framework can elevate the performance of a poorly-generalizing DNN model to a level that is statistically superior to or competitive with traditional machine learning models (e.g., Random Forests and SVR).

In the proposed methodology a DNN is first constructed and optimized to predict the cycle time of each job in the manufacturing system. Jobs with particularly accurate or inaccurate cycle time forecasts are then extracted into good and bad instances. For a new job if it is very similar to any bad instance its cycle time may be inaccurately predicted. The proposed model-agnostic Post-hoc Corrective Frameworks focus on such new jobs only. By measuring the similarity between these new jobs and good instances the proposed model-agnostic Post-hoc Corrective Frameworks attend to only similar good instances to fine-tune the attributes of the new job. After that the DNN is used without re-training to

predict the cycle time of the new job again to reduce the possible prediction error.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 introduces the proposed DNN and model-agnostic Post-hoc Corrective Framework approach. To evaluate the effectiveness of the proposed methodology, it has been applied to a real case in a wafer fab in Sect. 4. Section 5 is dedicated to a discussion of the experimental results. Finally, Sect. 6 concludes this study and puts forth some topics that deserve future investigation.

## 2 Literature review

Fan et al. [16] built a BiLSTM for online-retail sales forecasting and inserted a multimodal attention block between the encoder and decoder to boost accuracy. During decoding each predicted step drew attention to different portions of the history forming a tailored hidden representation for that future step. Their temporal attention scanned  $M$  historical periods and fused them using multimodal attention weights derived from interactions between prior hidden states and transformed content vectors. Following Fan et al. [16] we design an attention scheme that predicts cycle times of unseen (future) jobs by selectively attending to different parts of the learned (past) jobs. Whereas their multimodal attention emphasizes multiple segments of history, our approach considers the full historical record.

Shih et al. [27] used an LSTM with attention to forecast diverse time series solar output traffic occupancy electricity demand and exchange rates. In their setup LSTM hidden states were first passed through a CNN to filter salient patterns the CNN outputs were then fed back as additional inputs to the LSTM. This idea from Shih et al. [27] motivates us to tune what the DNN receives at its input layer to enhance predictive performance.

Adil et al. [29] modeled tourist arrivals with several BiLSTMs. Before modeling they applied STL (Loess) decomposition to split the series into trend seasonal and residual components and then predicted the trend and residual via separate BiLSTMs. They further added two attention layers to each BiLSTM - one over features and another over time steps-so that more informative attributes (or periods) were weighted more heavily prior to the BiLSTM layer. Although this design is straightforward to reproduce it alters the baseline BiLSTM optimality and demands retraining which in turn motivates our choice of a model-agnostic attention mechanism. Moreover, like Senoner et al. [41] and related work such attention can filter the input down to only the most important attributes however, there are many competing ways to measure importance and the most suitable criterion

may be application-specific (e.g., job cycle-time prediction here) and even vary across individual cases [29, 41].

Tian et al. [30] addressed wind-power forecasting with a GRU-based RNN augmented by a dual self-attention module that separately evaluates the relevance of inputs and hidden states using bias terms of the hidden features. The resulting adjustments to inputs and hidden states, applied before they enter the RNN, improved accuracy. Their strategy suggests that job attributes can be adaptively adjusted prior to feeding a DNN.

For new-product sales Ekambaram et al. [28] compared K-nearest neighbors with an RNN. Because product images influence demand they trained an encoder-decoder to compress image information into embeddings and merged those with temporal features (past sales and exogenous variables) before decoding with the RNN. They also employed self-attention to assign period-specific weights to temporal features based on the features themselves and the RNN hidden state ensuring that more informative attributes receive greater emphasis [28].

Most past research on using attention mechanisms for prediction has been on time series problems, which is not the case for the job cycle time prediction problem discussed in this study. In addition, hidden states in a DNN unlike those in an RNN or LSTM do not accumulate information so an attention mechanism based on them may not be valuable for adjusting inputs. Therefore, this study establishes a Post-hoc Corrective Framework specifically tailored to the job cycle time prediction problem. Table 1 summarizes the unique features of the proposed methodology compared with some existing methods.

### 3 Methodology

The proposed methodology constructs a DNN to predict the cycle time of a job and then applies two XAI technologies, namely Post-hoc Corrective Framework and SHAP analysis, jointly to improve the prediction performance. Among them, the Post-hoc Corrective Framework adjusts the

attributes of new jobs which cycle times may not be accurately predicted by attending similar old jobs whose cycle times were accurately predicted in the past. SHAP analysis is the most popular XAI technique used to evaluate the impact contribution or importance of each attribute of a job to its cycle time. Figure 1 provides a systemic diagram of the proposed methodology showing the role of the Post-hoc Corrective Framework.

Job attributes that have a greater impact on the cycle time forecast can be paid more attention to enhance the prediction performance [40]. It can be achieved in two ways. The first is to adjust the weights of the connections that such job attributes will pass through [29]. The second is to tune the values of such job attributes before they are inputted into the DNN [30]. The latter is simpler and is adopted in this study.

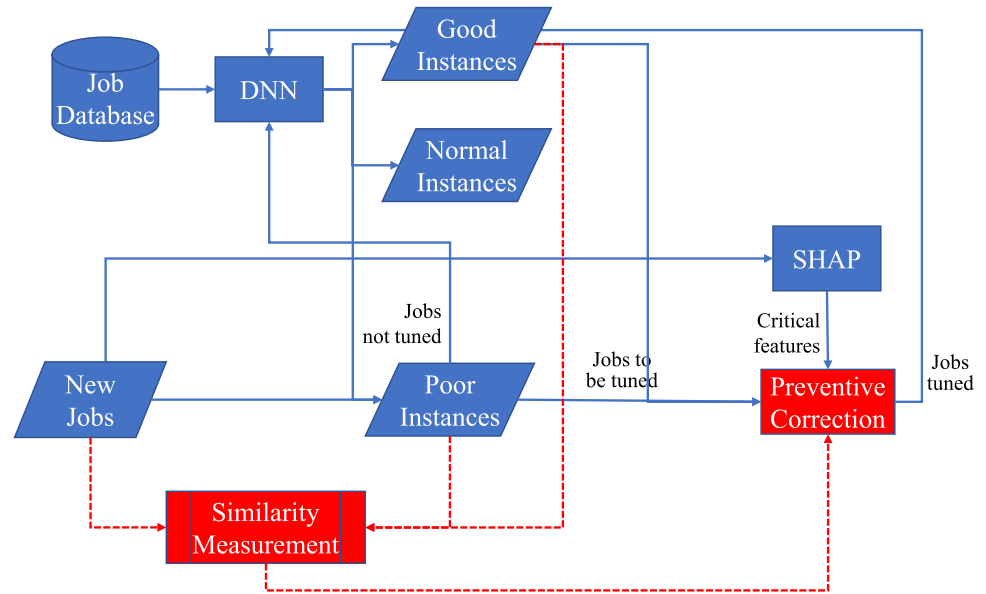
The proposed methodology comprises the following steps:

- Step 1. Build and train a DNN to predict the cycle times of jobs.
- Step 2. Identify old jobs (instances) which cycle times have been accurately and inaccurately predicted, respectively.
- Step 3. Evaluate the similarity between a new job and inaccurate instances which cycle times have been inaccurately predicted.
- Step 4. If the new job is very similar to any inaccurate instance go to Step 5; otherwise go to Step 8.
- Step 5. Evaluate the similarity between the new job and accurate instances.
- Step 6. If the new job is very similar to any accurate instance go to Step 7; otherwise go to Step 8.
- Step 7. Apply the global Post-hoc Corrective Framework to tune the attributes of the new job.
- Step 8. Apply the trained DNN to predict the cycle time of the new job.
- Step 9. Evaluate the effectiveness of the global Post-hoc Corrective Framework.
- Step 10. Conduct a SHAP analysis to determine the most important attributes for the new job.

**Table 1** Differences between the proposed methodology and some existing methods

| Method                | Prediction problem type   | Prediction method  | Attention/correction mechanism type       | Attention/correction mechanism parameters | Part of historical data attended |
|-----------------------|---------------------------|--------------------|-------------------------------------------|-------------------------------------------|----------------------------------|
| Fan et al. [16]       | Time series               | BiLSTM             | Global, soft, model-specific              | Hidden states, inputs                     | Multiple parts                   |
| Shih et al. [27]      | Time series               | LSTM, CNN          | Global, soft, model-specific              | Hidden states, inputs                     | Single part                      |
| Adil et al. [29]      | Time series               | BiLSTM             | Local, soft, model-specific               | Hidden states, inputs                     | Single part                      |
| Tian et al. [30]      | Time series               | GRU neural network | Global, soft, model-specific              | Hidden states, inputs                     | Single part                      |
| Ekambaram et al. [28] | Time series               | RNN                | Global, soft, model-specific              | Hidden states, inputs                     | Single part                      |
| Proposed methodology  | Input–output relationship | DNN                | Global, local, hard, soft, model-agnostic | Inputs                                    | All/Good instances               |

**Fig. 1** Systemic diagram of the proposed methodology



Step 11. Apply the local Post-hoc Corrective Framework to tune the attributes of the new job.

Step 12. Apply the trained DNN to predict the cycle time of the new job.

Step 13. Evaluate the effectiveness of the local Post-hoc Corrective Framework.

### 3.1 Step 1. Building and training a DNN to predict job cycle times

In the proposed methodology at first a DNN is built to predict the cycle time of a job. The DNN is a feedforward neural network (FNN) with  $V$  hidden layers aiming to minimize the root mean squared error (RMSE) in predicting the cycle times of jobs in the training data ( $J_{Tr}$ ):

$$\text{Min } RMSE_1 = \frac{1}{n} \sum_{j=1}^n (a_j - o_j)^2 \quad (1)$$

s.t.

$$o_j = DNN(x_j) = \frac{1}{1 + e^{-(\sum_{l_v=1}^{L_v} (w_{l_v}^o h_{jl_v}^{(v)}) - \theta^o)}}; j = 1 \sim n \quad (2)$$

$$h_{jl_v}^{(V)} = \frac{1}{1 + e^{-(\sum_{l_v=1}^{L_v} (w_{l_v}^{(V)} h_{jl_{v-1}}^{(V-1)}) - \theta_{l_v}^{(V)})}}; j = 1 \sim n; l_v = 1 \sim L_V \quad (3)$$

...

$$h_{jl_1}^{(1)} = \frac{1}{1 + e^{-(\sum_{m=1}^M (w_{ml_1}^{(1)} x_{jm}) - \theta_{l_1}^{(1)})}}; j = 1 \sim n; l_1 = 1 \sim L_1 \quad (4)$$

$$w_{ml_v}^{(v)}, \theta_{l_v}^{(v)}, w_{l_v}^o, \theta^o \in \mathbf{R}; m = 1 \sim M; l_v = 1 \sim L_v; v = 1 \sim V \quad (5)$$

where  $a_j$  and  $o_j$  denote the cycle time (actual value) and cycle time forecast of job  $j$ ;  $j = 1 \sim n$ .  $x_{jm}$  is the  $m$ -th attribute of job  $j$ ;  $m = 1 \sim M$ .  $h_{jl_v}^{(v)}$  is the output from node  $l_v$  in the  $v$  hidden layer.  $w_{ml_1}^{(1)}, w_{l_{v-1}l_v}^{(v)}, w_{l_v}^o$  indicate the connection weights between the input layer and the first hidden layer between hidden layers  $l_v$  and  $l_{v-1}$ , and between the last hidden layer and the output layer respectively. The most common algorithm for training the DNN is the Levenberg–Marquardt (LM) algorithm [42]. The prediction performance can be evaluated by applying the trained DNN to test data ( $J_{Te}$ ).

### 3.2 Step 2. Identifying accurate and inaccurate instances

The cycle time of a job has been accurately predicted if the absolute percentage error (APE) between its cycle time forecast and actual value is less than an accuracy threshold  $\xi$ :

$$APE_j = \frac{|a_j - o_j|}{a_j} \cdot 100\% < \xi \quad (6)$$

Such jobs are called accurate instances and put in a set  $J_a$ . Conversely if the APE is higher than another accuracy threshold  $\zeta$  the cycle time of the job has been inaccurately predicted:

$$APE_j \geq \zeta \quad (7)$$

These jobs are called inaccurate instances and put in another set  $J_b$ .

### 3.3 Step 3. Identifying new (unlearned) jobs that may be inaccurately predicted

The cycle time forecast of an unlearned job (i.e., a job in  $J_{Te}$ ) may be inaccurate if its attributes are very similar to those of any instance in  $J_b$ . For this purpose the (Euclidean) distance between two jobs are measured based on their attributes as

$$dist_1(j_1, j_2) = \sqrt{\sum_{m=1}^M \left( \frac{x_{j_1 m} - x_{j_2 m}}{\max_j x_{jm} - \min_j x_{jm}} \right)^2} \quad (8)$$

Then their similarity can be evaluated as

$$sim_1(j_1, j_2) = 2 - \frac{2}{1 + e^{-dist_1(j_1, j_2) \cdot c_1}} \quad (9)$$

which is derived from a tangent sigmoid function.  $c_1$  is a shaping constant.  $0 \leq sim_1(j_1, j_2) \leq 1$ . The effects of  $c_1$  on  $sim_1(j_1, j_2)$  is illustrated in Fig. 2. As the distance between two jobs increases their similarity decreases rapidly. In this way only very similar instances become references for a job.

The cycle time of a job  $j$  in  $J_{Te}$  may be incorrectly predicted if

$$sim_1(j_2, j) \geq s_2; j_2 \in J_b, j \in J_{Te} \quad (10)$$

$s_2$  is a similarity threshold;  $s_2 \in [0, 1]$ . Such jobs are put in a set  $J_c$ .

### 3.4 Step 4. Applying the global post-hoc corrective framework

For each job in  $J_c$  its attribute values are adjusted by referencing the close accurate instances in  $J_a$  as

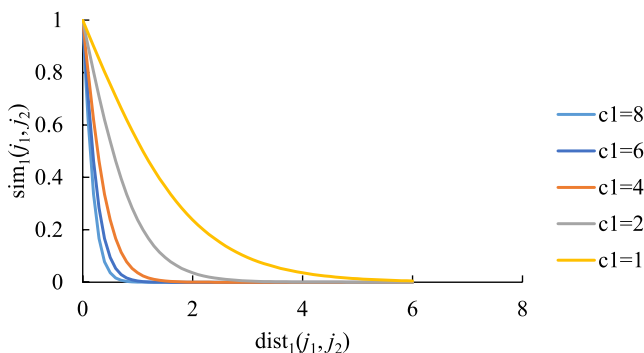


Fig. 2 Effects of  $c_1$  on  $sim_1(j_1, j_2)$

$$x_{jm}' = \frac{x_{jm} + \sum_{\substack{j_1 \in J_a \\ sim_1(j_1, j) \geq s_1}} (sim_1(j_1, j) \cdot x_{j_1 m})}{1 + \sum_{\substack{j_1 \in J_a \\ sim_1(j_1, j) \geq s_1}} sim_1(j_1, j)} \quad \forall j \in J_c \quad (11)$$

$s_1$  is another similarity threshold within  $[0, 1]$ . The modified attributes are inputted into the DNN to generate a new cycle time forecast  $o_j'$  for job  $j$ :

$$o_j' = DNN(x_j') \quad \forall a \quad (12)$$

### 3.5 Step 5. Evaluating the effectiveness of the global Post-hoc corrective framework

The effectiveness of the global Post-hoc Corrective Framework can be evaluated by comparing the RMSEs of job cycle time prediction before and after applying the global Post-hoc Corrective Framework:

$$Effectiveness_{Attention} = \frac{\sqrt{\sum_{j \in J_{Te}} (a_j - o_j)^2}}{|J_{Te}|} - \frac{\sqrt{\sum_{j \in J_{Te}-J_c} (a_j - o_j)^2 + \sum_{j \in J_c} (a_j - o_j')^2}}{|J_{Te}|} \quad (13)$$

### 3.6 Step 6. Conducting a SHAP analysis for attribute importance evaluation

The stability of the SHAP values is critical to the success of this framework. To address concerns regarding the potential sensitivity of SHAP values we conducted a robustness analysis across four-fold cross-validation. We observed that the top four ranked attributes (job size, fab WIP, bottleneck queue length and fab utilization) maintained a high degree of consistency across different training and testing splits. This consistency indicates that the selected important features are not spurious artifacts of a single model run but are stable key attributes with a sustained influence on the predictive outcome thereby demonstrating the robustness of our approach.

The SHAP value is defined as the weighted average of the marginal contributions of all possible coalitions  $|F|!$  as [43]:

$$\varphi_m(f) = \sum_{\{S \subseteq F\} \setminus \{m\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \cdot [f(x_{S \cup \{m\}}) - f(x_S)] \quad (14)$$

where  $\varphi_m(f)$  is the weighted average Shapley value that feature  $m$  provides in the context of all coalitions that exclude feature  $m$ .  $F$  is the set of all features;  $S$  is a subset (i.e., coalition) of  $F$ ;  $f(x_{S \cup \{m\}})$  is the model prediction considering feature  $m$ , while  $f(x_S)$  is the model prediction without considering feature  $m$ .

### 3.7 Step 7. Applying the local post-hoc corrective framework

In the local Post-hoc Corrective Framework the distance between two jobs is measured by considering only important attributes with high absolute SHAP values:

$$dist_2(j_1, j_2) = \sqrt{\sum_{|SHAP_m| \geq \varphi} \left( \frac{x_{j_1 m} - x_{j_2 m}}{\max_j x_{j m} - \min_j x_{j m}} \right)^2} \quad (15)$$

$\varphi$  is the SHAP threshold. Then their similarity is evaluated as

$$sim_2(j_1, j_2) = 2 - \frac{2}{1 + e^{-dist_2(j_1, j_2) \cdot c_2}} \quad (16)$$

$c_2$  is another shaping constant.  $0 \leq sim(j_1, j_2) \leq 1$ . The cycle time of a job  $j$  in  $J_{Te}$  may be incorrectly predicted if

$$sim_2(j_2, j) \geq s_3; j_2 \in J_b, j \in J_{Te} a \quad (17)$$

$s_3$  is another similarity threshold;  $s_3 \in [0, 1]$ . Such jobs are put in a set  $J_d$ .

Subsequently the attribute values of each job in  $J_d$  are tuned by referencing the accurate instances in  $J_a$  according to the following equations:

$$x_{jm} + \sum_{\substack{j_1 \in J_a \\ sim_2(j_1, j) \geq s_3}} \frac{x_{j_1 m}}{1 + \sum_{\substack{j_1 \in J_a \\ sim_2(j_1, j) \geq s_3 \\ \forall j \in J_d, |SHAP_m| \geq \varphi}} sim_2(j_1, j)} = \quad (18)$$

The tuned attribute values are inputted into the DNN to generate a new cycle time forecast  $o_j''$  for job  $j$ :

$$o_j'' = DNN(x_j'') \forall j \in J_d \quad (19)$$

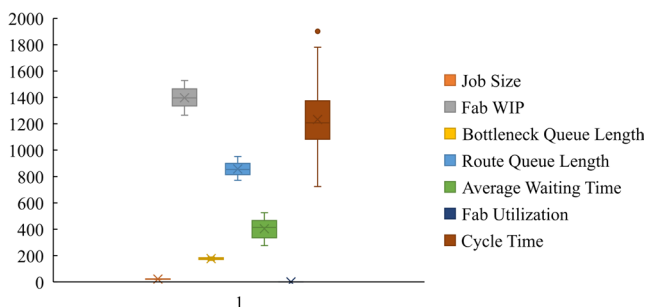


Fig. 3 Box-and-whisker plot for performing a pre-model analysis

### 3.8 Step 8. Evaluating the effectiveness of the local post-hoc corrective framework

The effectiveness of the local Post-hoc Corrective Framework is evaluated by comparing the RMSEs before and after applying the local Post-Hoc Corrective Framework:

$$Effectiveness_{SHAP+Attention} = \frac{\sqrt{\sum_{j \in J_{Te}} (a_j - o_j)^2}}{|J_{Te}|} - \frac{\sqrt{\sum_{j \in J_{Te}-J_d} (a_j - o_j)^2 + \sum_{j \in J_d} (a_j - o_j'')^2}}{|J_{Te}|} \quad (20)$$

## 4 Case Study

### 4.1 Background

The data for this study were sourced from the Manufacturing Execution System (MES) of a real-world, high-volume semiconductor wafer fabrication facility. In this manufacturing context, a job refers to a lot of 25 silicon wafers that undergoes several hundred distinct processing steps over multiple weeks. Our research goal is to predict the total job cycle time (from release to completion), which is a system-level problem heavily influenced by factory-wide queuing dynamics rather than specific machine-level parameters. Therefore, the dataset comprises 300 jobs, characterized by the high-level system-state and job-specific attributes (e.g., Job Size, Fab WIP) that are standard variables used in industry for such system-level forecasting.

To assess the effectiveness of the proposed methodology in improving the accuracy of job cycle time prediction, the real case investigated in Chen and Lin [11] was adopted, in which the cycle times ( $o_j$ ) of 300 jobs in a wafer fab were predicted based on their six attributes and production conditions when these jobs were released into the wafer fab: job size ( $x_{j1}$ ), fab work-in-process (WIP) ( $x_{j2}$ ), the total queue length before bottlenecks ( $x_{j3}$ ), fab queue length ( $x_{j4}$ ), the average waiting time of recently completed jobs ( $x_{j5}$ ), and fab utilization ( $x_{j6}$ ). A box-and-whisker plot (see Fig. 3) was drawn to perform a pre-model analysis. The data contained very few outliers and was not skewed. Therefore, no specific preprocessing was performed. In addition, a violin plot (see Fig. 4) was also drawn which showed the frequency of each job attribute in taking various values.

### 4.2 Application of DNN

A DNN (i.e. an FNN with multiple hidden layers) was constructed to predict the cycle time of each job. fourfold cross-validation was adopted to verify the effectiveness of the DNN. Namely, the training data contained 225 jobs, while test data had remaining 75 jobs. Therefore,  $|J_{Tr}| = 225$  and



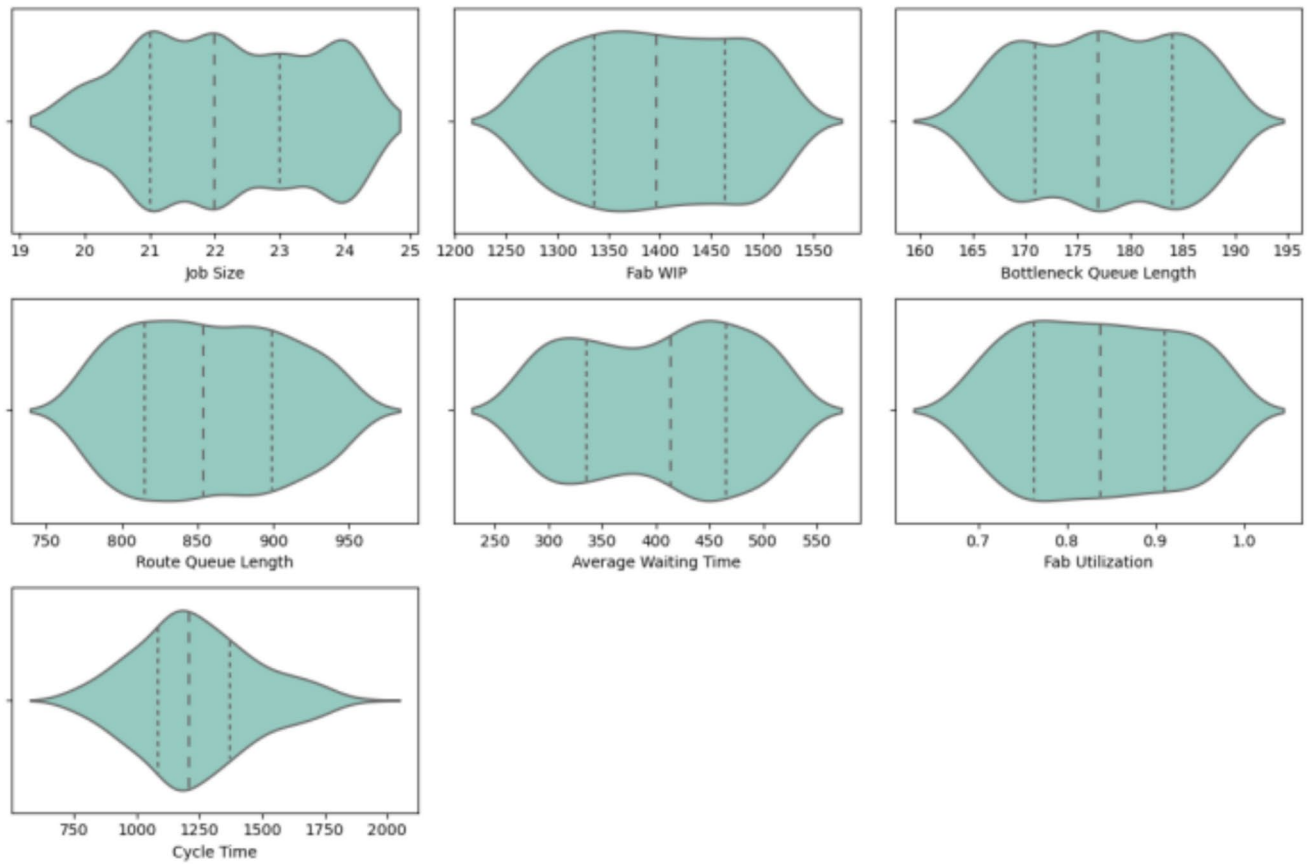


Fig. 4 Violin plot

$|J_{Te}| = 75$ ;  $J_{Tr} \cap J_{Te} = \emptyset$ . The prediction accuracy was measured in terms of mean squared error (MSE) mean absolute percentage error (MAPE) RMSE and  $R^2$ . The first three performance measures were the lower the better while the last performance measure was the higher the better. The LM algorithm [42] was applied to train the DNN. The maximum number of epochs was 50000. The target for the network convergence was RMSE less than 100 h. The configuration of the DNN was optimized by trying two to four hidden layers with one to twelve nodes in each hidden layer [44, 45]. For each configuration the results of ten iterations were averaged. According to the parametric analysis results the optimal configuration of the DNN was determined to have two hidden layers with 6 and 5 neurons respectively (see Fig. 5).

After training the DNN was applied to test data, for which the prediction accuracy was evaluated as

$$\begin{aligned} \text{MAE} &= 158.5(\text{hrs}) \\ \text{MAPE} &= 12.5\% \\ \text{RMSE} &= 205.1(\text{hrs}) \\ R^2 &= 0.24 \end{aligned}$$

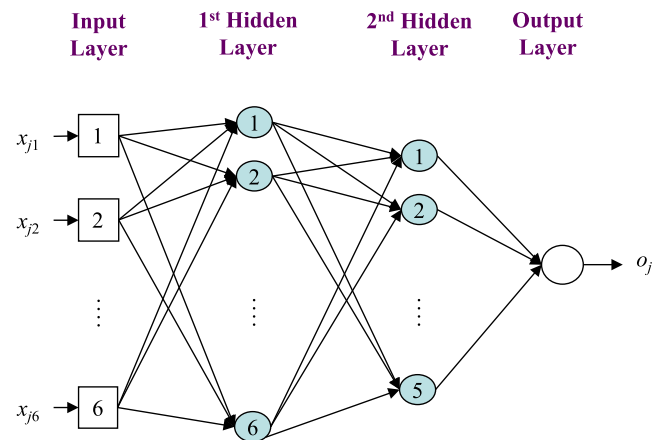
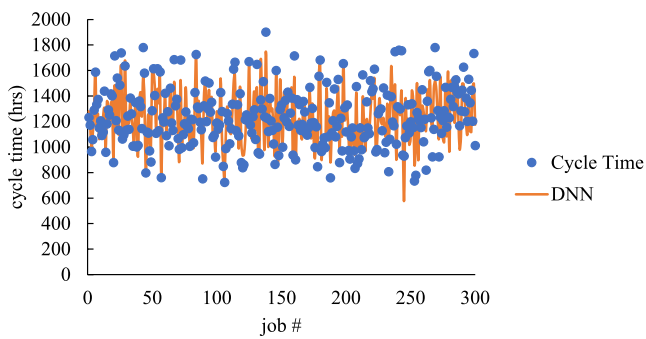


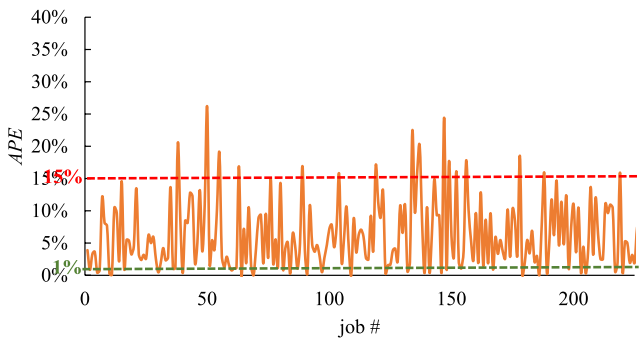
Fig. 5 Optimal configuration of the DNN

### 4.3 Application of the global Post-hoc corrective framework

Taking the first fold of the training data in the first iteration as an example, i.e.,  $J_{Tr} = \{j_1 \sim j_{225}\}$  and  $J_{Te} = \{j_{226} \sim j_{300}\}$ . The prediction results are summarized in Fig. 6 with the following accuracy:



**Fig. 6** Prediction results of the first fold in the first iteration

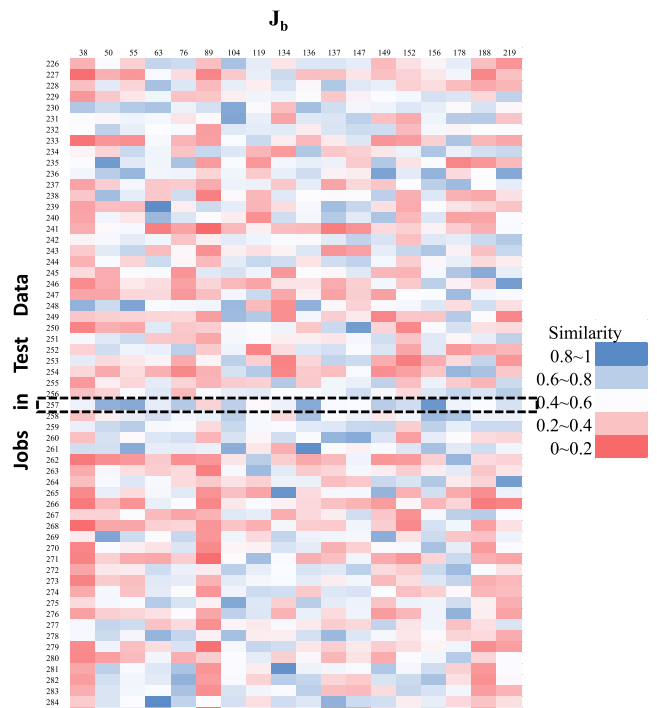


**Fig. 7** APEs of jobs in  $J_{Tr}$

$$\begin{aligned} \text{MAE} &= 167.8(\text{hrs}) \\ \text{MAPE} &= 13.7\% \\ \text{RMSE} &= 195.9(\text{hrs}) \\ R^2 &= 0.31 \end{aligned}$$

Subsequently, the APE of each job in  $J_{Tr}$  was calculated to determine whether its cycle time prediction was very accurate or inaccurate (see Fig. 7) for which the accuracy thresholds  $\xi$  and  $\zeta$  were specified as 1% and 15%, respectively. As a result the numbers of instances which cycle time forecasts were (very) accurate and inaccurate were 28 and 18 respectively. Namely,  $|J_a| = 28$  and  $|J_b| = 18$  (see Table 2).

The similarity between each job in  $J_{Te}$  and every inaccurate instance in  $J_b$  was evaluated, for which  $c_1 = 6$ . The evaluation results are summarized with a heatmap in Fig. 8. If the row for a job in  $J_{Te}$  contains many close-to-blue grids the cycle time of the job may be incorrectly predicted e.g., job #257 in Fig. 8. By setting the similarity threshold  $s_2$  to 0.625, the number of such jobs was 9. Namely,  $J_c$  had nine jobs which attributes were to be fine-tuned.



**Fig. 8** Similarity between each job in test data and every inaccurate instance in  $J_b$

Subsequently, the attributes of each job in  $J_c$  were tuned with reference to the close accurate instances in  $J_a$ , for which the similarity threshold  $s_1$  was set to 0.5. Then the DNN was applied to forecast the cycle times of jobs in  $J_{Te} - J_c$  based on the original attribute values and jobs in  $J_c$  based on the tuned attribute values. For jobs with cycle times changed the cycle time forecasts based on the original attribute values ( $o_j$ ) and those based on the tuned attribute values ( $o_j'$ ) are compared in Table 3. By tuning the attribute values of such jobs beforehand most of their cycle times could be more accurately forecasted. The overall prediction performance changed to

$$\begin{aligned} \text{MAE} &= 166.3(\text{hrs}) \\ \text{MAPE} &= 13.6\% \\ \text{RMSE} &= 194.6(\text{hrs}) \\ R^2 &= 0.32 \end{aligned}$$

$\text{Effectiveness}_{\text{Attention}} = 195.9 - 194.6 = 1.3(\text{hrs})$  which was the effectiveness of the global Post-hoc Corrective Framework in only a single fold and iteration without optimization.

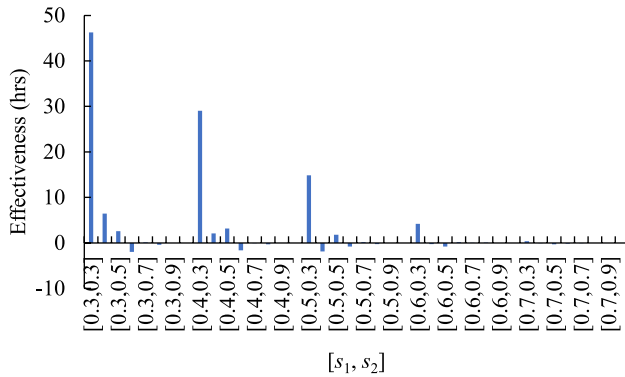
**Table 2** Jobs with very accurate and inaccurate cycle time forecasts

| Job Set | Job #                                                                                                                      |
|---------|----------------------------------------------------------------------------------------------------------------------------|
| $J_a$   | 2, 5, 6, 10, 11, 30, 40, 60, 64, 69, 84, 91, 97, 109, 123, 132, 139, 141, 146, 148, 167, 179, 186, 189, 203, 205, 217, 220 |
| $J_b$   | 38, 50, 55, 63, 76, 89, 104, 119, 134, 136, 137, 147, 149, 152, 156, 178, 188, 219, 207, 224                               |



**Table 3** Comparison of prediction results

| $j$ | $a_j$ (hrs) | $o_j$ (hrs) | $o_j'$ (hrs) | Improvement (hrs) |
|-----|-------------|-------------|--------------|-------------------|
| 257 | 1208        | 1435.8      | 1396.3       | 39.5              |
| 261 | 1219        | 1202.1      | 1176.7       | −25.4             |
| 288 | 1167        | 980.6       | 1030.0       | 49.5              |
| 294 | 1201        | 1100.9      | 1148.5       | 47.6              |

**Fig. 9** Performance of the global post-hoc corrective framework with various values of  $s_1$  and  $s_2$  (fold-fold cross-validation, 10 iterations)

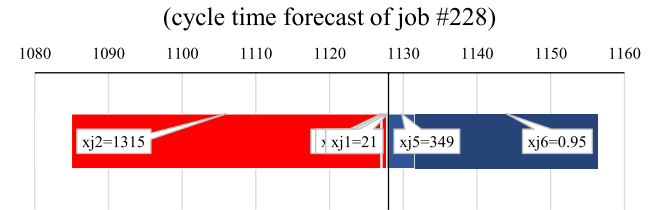
#### 4.4 Optimizing the effectiveness of the global post-hoc corrective framework

The values of the similarity thresholds  $s_1$  and  $s_2$  can be chosen to optimize the performance of the global Post-hoc Corrective Framework. For this purpose, various values of the two parameters were tried and the effectiveness of the global Post-hoc Corrective Framework was evaluated (by averaging the results of all four folds and ten iterations). The evaluation results are summarized in Fig. 9. Obviously by setting  $s_1$  and  $s_2$  to 0.3 and 0.3 respectively the performance of the global Post-hoc Corrective Framework was maximized which reduced RMSE from 205.1 h to 158.58 h with an improvement of 22.5%.

To ensure fair and systematic parameter selection, a grid search procedure was conducted to optimize the two similarity thresholds used in the global correction stage. Multiple combinations within [0.1, 0.9] were evaluated through four-fold cross-validation and the configuration yielding the lowest average RMSE was selected. The best-performing values (0.3 and 0.3) consistently provided stable convergence across iterations, confirming that the optimization process was not sensitive to initialization or data partitioning.

**Table 4** Importance of each job attribute in terms of the SHAP value

| $j$ | SHAP value |        |        |        |        |        |
|-----|------------|--------|--------|--------|--------|--------|
| 226 | −18.94     | −44.40 | −9.56  | −10.11 | −23.06 | −18.77 |
| 227 | 39.09      | 21.50  | 39.88  | 56.63  | 32.60  | 26.53  |
| 228 | −0.24      | −43.02 | −0.89  | −1.01  | 3.56   | 28.38  |
| ... |            |        |        |        |        |        |
| 300 | −3.31      | −14.82 | −26.84 | −55.89 | −81.96 | −73.85 |

**Fig. 10** Comparison of the importance levels of various attributes for job # 228

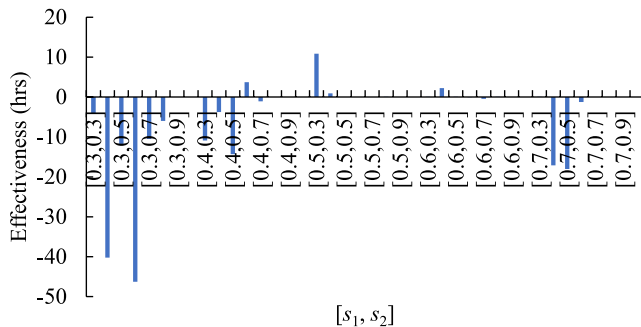
#### 4.5 Application of the local post-hoc corrective framework

Subsequently, a SHAP analysis was performed to evaluate the importance of each attribute for every job. The results are summarized in Table 4. Taking job #228 as an example. The importance levels (i.e., SHAP values) of various attributes are compared in Fig. 10. To this job the most important attributes included  $x_{j2}$ ,  $x_{j6}$  and  $x_{j5}$ . After averaging the absolute SHAP values of all jobs, the most important attributes overall were  $x_{j5}$ ,  $x_{j6}$ ,  $x_{j4}$  and  $x_{j3}$ . Therefore, by setting the SHAP threshold  $\phi$  to 25, the local Post-hoc Corrective Framework considered only the four attributes in measuring distances and evaluating similarity. Fewer attributes elevated similarity. Therefore, the similarity thresholds  $s_1$  and  $s_2$  were increased to 0.625 and 0.75 respectively.  $|J_a| = 24$  and  $|J_b| = 21$ . The attributes of 9 jobs were fine-tuned and their cycle times were forecasted based on the tuned attributes. For these jobs the cycle time forecasts before and after applying the local Post-hoc Corrective Framework are compared in Table 5. In addition, the prediction performance after applying the local Post-hoc Corrective Framework was:

$$\begin{aligned}
 \text{MAE} &= 141.86(\text{hrs}) \\
 \text{MAPE} &= 11.0\% \\
 \text{RMSE} &= 178.0(\text{hr}) \\
 R^2 &= 0.43
 \end{aligned}$$

**Table 5** Comparison of prediction results

| $j$ | $a_j$ (hrs) | $o_j$ (hrs) | $o_j^l$ (hrs) | Improvement (hrs) |
|-----|-------------|-------------|---------------|-------------------|
| 236 | 1197        | 1474.55     | 1432.11       | 42.44             |
| 237 | 1167        | 1205.68     | 1179.37       | 26.31             |
| 242 | 1300        | 1130.99     | 1204.06       | 73.07             |
| 251 | 1340        | 1309        | 1290.09       | −18.90            |
| 259 | 1450        | 1230.2      | 1204.15       | −26.05            |
| 260 | 1042        | 1123.97     | 1092.92       | 31.05             |
| 265 | 1602        | 1464.76     | 1788.65       | −49.41            |
| 276 | 1285        | 1176.57     | 1199.35       | 22.77             |
| 299 | 1734        | 1211.48     | 1696.06       | 484.58            |

**Fig. 11** Performance of the local post-hoc corrective framework with various values of  $s_1$  and  $s_2$ 

Furthermore,  $Effectiveness_{SHAP+Attention} = 10.65(\text{hrs})$ .

#### 4.6 Optimizing the performance of the local post-hoc corrective framework

The values of the similarity thresholds  $s_1$  and  $s_2$  can be optimized to enhance the prediction performance using the local Post-hoc Corrective Framework. To this end various values of the two parameters were enumerated and tried to assess the effectiveness of the local Post-hoc Corrective Framework. The evaluation results are summarized in Fig. 11. By setting  $s_1$  and  $s_2$  to 0.5 and 0.3 respectively the performance of the local Post-hoc Corrective Framework was maximized which reduced RMSE from 205.1 h to 194.26 h with an improvement of 5.3%.

Similar to the global correction phase, the local framework's parameters were optimized through a systematic grid search across the same range of threshold values. The pair (0.5, 0.3) achieved the best balance between accuracy improvement and model robustness. Furthermore, the

SHAP cutoff value was set to 25 after examining the stability of feature importance across cross-validation folds, ensuring that only attributes with consistently high explanatory power were retained for local correction.

## 5 Discussion

Perhaps the most important takeaway from our results is how a targeted post-hoc correction can rescue a deep learning model that otherwise struggles. Our initial DNN was not very effective on its own with a low  $R^2$  of just 0.24. But after applying our framework we saw a dramatic 22.5% drop in RMSE - an improvement that a paired-samples  $t$ -test confirmed was statistically significant ( $p < 0.01$ ). What's more this wasn't just an internal improvement. The corrected DNN with an RMSE of 158.5 h actually ended up outperforming a standard Random Forests model (RMSE = 194.3 h), which is widely regarded as a strong baseline for nonlinear regression in complex prediction tasks [48].

The local Post-hoc Corrective Framework in this study was designed to enhance efficiency by reducing the number of attributes considered its performance appears less stable than the global correction. This suggests that for this particular manufacturing data set the complex interplay between all attributes may be more critical for accurate predictions than the isolated impact of the few most important features identified by SHAP. When the similarity thresholds are set too low the mechanism may reference less relevant instances leading to unstable performance. Conversely when the thresholds are too high there are too few instances to reference and the correction mechanism becomes ineffective. Nevertheless, the established framework rarely worsened

**Table 6** Prediction performances using various methods

| Method                                    | MAE (hrs) | MAPE  | RMSE (hrs) | $R^2$ |
|-------------------------------------------|-----------|-------|------------|-------|
| DNN                                       | 158.5     | 12.5% | 205.1      | 0.24  |
| DNN + SHAP                                | 178.9     | 14.6% | 221.5      | 0.11  |
| DNN + SHAP + hybrid preventive correction | 136.3     | 10.8% | 180.4      | 0.42  |
| DNN + SHAP + local preventive correction  | 155.9     | 12.4% | 194.3      | 0.31  |
| DNN + global preventive correction        | 137.8     | 10.7% | 158.6      | 0.49  |

prediction performance and its impact was negligible which supports the robustness of the proposed methodology and is consistent with findings in recent XAI studies showing that post-hoc, model-agnostic explanations can be integrated with predictive models without degrading their reliability [46, 47, 49]. Furthermore compared to the common practice of only considering attributes with high SHAP values for prediction our framework demonstrated superior performance, suggesting that while SHAP values are derived from the DNN feeding them back to improve the model is not a straightforward process. Table 6 compares the prediction performance using various methods.

## 6 Conclusions

This study establishes a SHAP-driven post-hoc correction framework to enhance the effectiveness of a DNN in job cycle time prediction. Unlike existing mechanisms for time series forecasting this framework is tailored for input–output relationships. It does not correct all new jobs but selectively targets only those with potentially inaccurate predictions. Our SHAP analysis also demonstrated the feasibility of locally considering more important attributes for correction.

The proposed methodology was validated in a real-world case from a wafer fab with 300 jobs. Experimental results show that the framework successfully improved the predictive accuracy of the DNN model by up to 22.5%, and this improvement was found to be statistically significant. Moreover, comparative analysis against classic benchmark models like Random Forests confirmed the framework's effectiveness.

Future research can explore applying this framework to larger more diverse datasets to further verify its scalability. Additionally, other types of similarity evaluation functions can be investigated to replace the current approach. Finally, the systematic selection of parameter values within the framework to achieve optimal improvement is a topic that warrants further investigation.

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**Data availability** Data will be made available on request.

## Declarations

**Ethical approval** Not required.

**Conflicts of interest** The author declares that there is no conflict of interest regarding the publication of this article.

**Guarantor** Not required.

## References

1. Chen TCT (2022) Cycle time prediction and output projection. In: Chien CF, Hsieh SY, Chen TCT (eds) *Production Planning and Control in Semiconductor Manufacturing: Big Data Analytics and Industry 4.0 Applications*. Springer, Cham, pp 41–62.
2. Chen T (2015) An efficient and effective fuzzy collaborative intelligence approach for cycle time estimation in wafer fabrication. *Int J Intell Syst* 30(6):620–650
3. Lee GM, Gao X (2021) A hybrid approach combining fuzzy C-means-based genetic algorithm and machine learning for predicting job cycle times for semiconductor manufacturing. *Appl Sci* 11(16):7428
4. Chen T, Wu HC (2017) A new cloud computing method for establishing asymmetric cycle time intervals in a wafer fabrication factory. *J Intell Manuf* 28:1095–1107
5. Wang J, Zhang J, Wang X (2018) A data driven cycle time prediction with feature selection in a semiconductor wafer fabrication system. *IEEE Trans Semicond Manuf* 31(1):173–182
6. Wang J, Zhang J, Wang X (2018) Bilateral LSTM: a two-dimensional long short-term memory model with multiply memory units for short-term cycle time forecasting in re-entrant manufacturing systems. *IEEE Trans Ind Inform* 14(2):748–758
7. Wang J, Yang J, Zhang J, Wang X, Zhang W (2018) Big data driven cycle time parallel prediction for production planning in wafer manufacturing. *Enterp Inf Syst* 12(6):714–732
8. Fang W, Guo Y, Liao W, Ramani K, Huang S (2020) Big data driven jobs remaining time prediction in discrete manufacturing system: a deep learning-based approach. *Int J Prod Res* 58(9):2751–2766
9. Chen TCT, Wang YC (2021) Fuzzy dynamic-prioritization agent-based system for forecasting job cycle time in a wafer fabrication plant. *Complex Intell Syst* 7:2141–2154
10. Wang YC, Tsai HR, Chen T (2021) A selectively fuzzified back propagation network approach for precisely estimating the cycle time range in wafer fabrication. *Mathematics* 9(12):1430
11. Chen TCT, Lin YC (2022) Fuzzified deep neural network ensemble approach for estimating cycle time range. *Appl Soft Comput* 130:109697
12. Chen T, Wang YC (2022) Hybrid big data analytics and Industry 4.0 approach to projecting cycle time ranges. *Int J Adv Manuf Technol* 120(1–2):279–295
13. Sun C, Dominguez-Caballero J, Ward R, Ayvar-Soberanis S, Curtis D (2022) Machining cycle time prediction: data-driven modelling of machine tool feedrate behavior with neural networks. *Robot Comput Integr Manuf* 75:102293
14. Gao P, Wang J, Zhong R, Zhang J (2023) Neuron synergy based explainable neural network for manufacturing cycle time forecasting. *J Manuf Syst* 71:695–706
15. Wang YC, Chen T, Chiu MC (2023) An explainable deep-learning approach for job cycle time prediction. *Decision Analytics Journal* 6:100153
16. Fan C, Zhang Y, Pan Y, Li X, Zhang C, Yuan R, Huang H (2019) Multi-horizon time series forecasting with temporal attention learning. In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '19)*, Anchorage, AK, USA, pp 2527–2535.
17. Hasanpour Zaryabi E, Moradi L, Kalantar B, Ueda N, Halin AA (2022) Unboxing the black box of attention mechanisms in remote sensing big data using XAI. *Remote Sens* 14(24):6254
18. Ibrahim SM, Ansari SS, Hasan SD (2023) Towards white box modeling of compressive strength of sustainable ternary cement concrete using explainable artificial intelligence (XAI). *Appl Soft Comput* 149:110997

19. Hu Q, Gois FNB, Costa R, Zhang L, Yin L, Magaia N, de Albuquerque VHC (2022) Explainable artificial intelligence-based edge fuzzy images for COVID-19 detection and identification. *Appl Soft Comput* 123:108966
20. Chen TCT, Lin CW, Lin YC (2024) A fuzzy collaborative forecasting approach based on XAI applications for cycle time range estimation. *Appl Soft Comput* 151:111122
21. Messner W (2023) From black box to clear box: a hypothesis testing framework for scalar regression problems using deep artificial neural networks. *Appl Soft Comput* 146:110729
22. Chen TCT, Wu HC, Chiu MC (2023) A deep neural network with modified random forest incremental interpretation approach for diagnosing diabetes in smart healthcare. *Appl Soft Comput* 152:111183
23. Jiang H, Lu Y, Zhang D, Shi Y, Wang J (2024) Deep learning-based fusion networks with high-order attention mechanism for 3D object detection in autonomous driving scenarios. *Appl Soft Comput*. <https://doi.org/10.1016/j.asoc.2024.111253>
24. Gao D, Yang W, Li P, Liu S, Liu T, Wang M, Zhang Y (2024) A multiscale feature fusion network based on attention mechanism for motor imagery EEG decoding. *Appl Soft Comput* 151:111129
25. Yang ZB, Zhang JP, Zhao ZB, Zhai Z, Chen XF (2020) Interpreting network knowledge with attention mechanism for bearing fault diagnosis. *Appl Soft Comput* 97:106829
26. Wu P, Li X, Ling C, Ding S, Shen S (2021) Sentiment classification using attention mechanism and bidirectional long short-term memory network. *Appl Soft Comput* 112:107792
27. Shih SY, Sun FK, Lee HY (2019) Temporal pattern attention for multivariate time series forecasting. *Mach Learn* 108:1421–1441
28. Ekambaram V, Manglik K, Mukherjee S, Sajja SSK, Dwivedi S, Raykar V (2020) Attention-based multi-modal new product sales time-series forecasting. In: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '20)*, Virtual Event, pp 3110–3118
29. Adil M, Wu JZ, Chakraborty RK, Alahmadi A, Ansari MF, Ryan MJ (2021) Attention-based STL-BiLSTM network to forecast tourist arrival. *Processes* 9(10):1759
30. Tian C, Niu T, Wei W (2022) Developing a wind power forecasting system based on deep learning with attention mechanism. *Energy* 257:124750
31. Yu C, Yan G, Yu C, Mi X (2023) Attention mechanism is useful in spatio-temporal wind speed prediction: evidence from China. *Appl Soft Comput* 148:110864
32. Liu X, Zhou J (2024) Short-term wind power forecasting based on multivariate/multi-step LSTM with temporal feature attention mechanism. *Appl Soft Comput* 150:111050
33. Su L, Xiong L, Yang J (2023) Multi-Attn BLS: multi-head attention mechanism with broad learning system for chaotic time series prediction. *Appl Soft Comput* 132:109831
34. Dai Y, Zhou Q, Leng M, Yang X, Wang Y (2022) Improving the Bi-LSTM model with XGBoost and attention mechanism: a combined approach for short-term power load prediction. *Appl Soft Comput* 130:109632
35. Seth D, Seth N, Dhariwal P (2017) Application of value stream mapping (VSM) for lean and cycle time reduction in complex production environments: a case study. *Prod Plan Control* 28(5):398–419
36. Zhou Z, Rose O (2013) Cycle time variance minimization for WIP balance approaches in wafer fabs. In: *2013 Winter Simulation Conference (WSC)*, Washington, DC, USA, pp 3777–3788.
37. Chen T, Lin YC (2021) Enhancing the accuracy and precision of forecasting the productivity of a factory: a fuzzified feedforward neural network approach. *Complex Intell Syst* 7(5):2317–2327
38. Daroudi S, Kazemipoor H, Najafi E, Fallah M (2021) The minimum latency in location routing fuzzy inventory problem for perishable multi-product materials. *Applied Soft Computing* 110:107543
39. Chen T (2008) A fuzzy-neural and multiple-bucket approach for estimating lot cycle time in a wafer fab with dynamic product mix. *Comput Ind Eng* 55(2):423–438
40. Luong MT, Pham H, Manning CD (2015) Effective approaches to attention-based neural machine translation. Preprint at <https://arxiv.org/abs/1508.04025>
41. Senoner J, Netland T, Feuerriegel S (2022) Using explainable artificial intelligence to improve process quality: evidence from semiconductor manufacturing. *Manag Sci* 68(8):5704–5723
42. Nocedal J, Wright SJ (2006) *Numerical Optimization*, 2nd edn. Springer, New York.
43. Joseph LP, Joseph EA, Prasad R (2022) Explainable diabetes classification using hybrid Bayesian-optimized TabNet architecture. *Comput Biol Med* 151:106178
44. Stathakis D (2009) How many hidden layers and nodes? *Int J Remote Sens* 30(8):2133–2147
45. Panchal G, Ganatra A, Kosta YP, Panchal D (2011) Behaviour analysis of multilayer perceptrons with multiple hidden neurons and hidden layers. *Int J Comput Theor Eng* 3(2):332–337
46. Kumarakulasinghe NB, Blomberg T, Liu J, Leao AS, Papapetrou P (2020) Evaluating local interpretable model-agnostic explanations on clinical machine learning classification models. In: *2020 IEEE 33rd International Symposium on Computer-Based Medical Systems (CBMS)*, Rochester, MN, USA, pp 7–12
47. Kamath U, Liu J (2021) *Explainable Artificial Intelligence: An Introduction to Interpretable Machine Learning*. Springer, Cham.
48. Rodriguez-Galiano V, Sanchez-Castillo M, Chica-Olmo M, Chica-Rivas MJOR (2015) Machine learning predictive models for mineral prospectivity: an evaluation of neural networks, random forest, regression trees and support vector machines. *Ore Geol Rev* 71:804–818
49. Zafar MR, Khan N (2021) Deterministic local interpretable model-agnostic explanations for stable explainability. *Mach Learn Knowl Extr* 3(3):525–541

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